

SYDNEY TECHNICAL HIGH SCHOOL

(Celebrating 100 Years of Public Education)



YEAR 11 YEARLY EXAMINATION

PRELIMINARY HSC ASSESSMENT TASK 3

2011

Mathematics Extension 1

General Instructions

- Working time - 90 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Diagrams are not drawn to scale

Total marks - 66

- Attempt Questions 1 – 6
- All questions are of equal value
- Start each question on a new page

Name : _____

Teacher : _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	TOTAL

Question 1 (11 marks)

- (a) Express the following parametric equations in Cartesian form. 1

$$x = t + 1, y = t(t + 4)$$

- (b) Solve $\frac{x}{x+2} < 0$ 2

- (c) Find, correct to the nearest degree, the acute angle between the lines 2

$$2x - y + 1 = 0 \text{ and } 3x - 2y + 4 = 0$$

- (d) Write down the equation of a monic polynomial of degree 3 that is 2
an odd function and has a root at $x = 2$.

- (e) If α, β and γ are the roots of the equation $2x^3 - 4x^2 + 6x + 8 = 0$
find the value of

(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1

(iii) $\alpha\beta\gamma$ 1

(iv) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$ 1

Question 2 (11 marks) - Start a new page

- (a) Evaluate $\lim_{x \rightarrow \infty} \frac{x^2 - x - 2}{3x^2 - 3x + 1}$ 1
- (b) Solve the equation $2x = \sqrt{7x + 11}$ 3
- (c) If θ is acute and $\cos 2\theta = \frac{3}{4}$, find the exact value of $\cos \theta$. 2
- (d) Factorise fully the polynomial $2x^3 + 11x^2 + 17x + 6$ 3
- (e) Find the values of a and b if the solution of the inequality $|x - a| \leq b$ is given by $2 \leq x \leq 5$. 2

Question 3 (11 marks) - Start a new page

- (a) Find the values of A , B and C if 3

$$2x^2 + 3x - 4 \equiv A(x - 1)^2 + Bx(x - 2) + C(x - 1)$$

- (b) Find the value of k if the line $2x - y + 2 + k(x + 3y - 1) = 0$ 2

is parallel to the line $x + 2y + 4 = 0$.

- (c) Use the substitution $t = \tan \frac{x}{2}$ to solve the equation 4

$$2\sin x - 5\cos x = 2 \text{ for } 0^\circ \leq x \leq 360^\circ \text{ correct to the nearest degree.}$$

- (d) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

- (i) Show that the gradient of the chord PQ is given by $\frac{p+q}{2}$. 1

- (ii) Find the equation of the chord PQ . 1

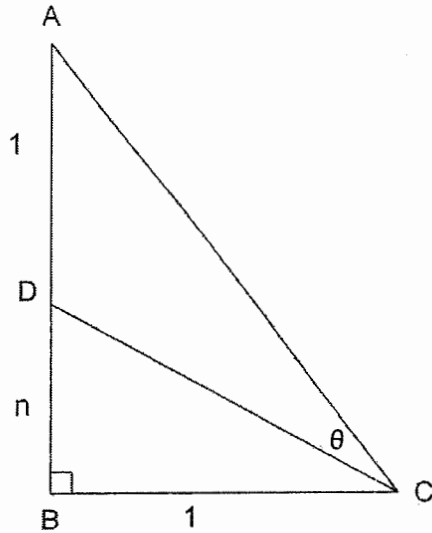
Question 4 (11 marks) - Start a new page

- (a) Find the vertex and focus of the parabola $y = \frac{1}{8}x^2 + x$ 2
- (b) Find the remainder when $x^3 + 4x - 4$ is divided by $x^2 - 1$ 2
- (c) Find the value, or values of k if the line $y = kx - 16$ 3
is a tangent to the parabola $y = x^2 - 2x$.
- (d) Solve the equation $2\sin 2\theta = 3\tan \theta$ for $0 \leq \theta \leq 2\pi$. 4
(Give your answers in radians)

Question 5 (11 marks) - Start a new page

(a)

3



In triangle ABC , $\text{angle } ABC = 90^\circ$, $AD = BC = 1$ unit and $BD = n$ units.

If $\text{angle } ACD = \theta$ find an expression for $\tan \theta$ in terms of n .

(b) Consider the variable point $P(4p, 2p^2)$ on the parabola $x^2 = 8y$.

(i) Prove that the equation of the tangent at P is $y = px - 2p^2$.

2

(ii) If the tangent at P meets the y axis at the point T ,

1

find the coordinates of T .

(iii) If S is the focus of the parabola $x^2 = 8y$, show that $TS = PS$.

2

(iv) If the point M divides the interval PT internally in the ratio 1:3,

3

find the locus of M as the point P varies.

Question 6 (11 marks) - Start a new page

- (a) Let A and B be the fixed points $(5, 6)$ and $(-1, 0)$ respectively and P be the variable point (x, y) .
- (i) Show that the locus of all points P , such that the distance from P to A is twice the distance from P to B , is a circle. 2
- (ii) Find the centre and radius of this circle. 2
- (b) The equation $x^3 + 6x^2 - x + m = 0$ has one root equal to the sum of the other two roots.
- (i) Find the value of m . 2
- (ii) Solve the equation. 2
- (c) Simplify $\frac{\tan 2\theta - \tan \theta}{\cot \theta + \tan 2\theta}$ 3

Teacher's Name:

Student's Name/N^o:

1. a. $t = x - 1$

$$\therefore y = (x-1)(x+3)$$

b. $\frac{x}{x+2} < 0$

$$-2 < x < 0$$

c. $m_1 = 2 \quad m_2 = \frac{3}{2}$

$$\tan \theta = \frac{2 - \frac{3}{2}}{1 + 2 \times \frac{3}{2}}$$

$$= \frac{\frac{1}{2}}{4}$$

$$= \frac{1}{8}$$

$$\therefore \theta = 7^\circ$$

d. $y = x(x-2)(x+2)$

e. i) $\alpha + \beta + \gamma = 2$

ii) $\alpha\beta + \alpha\gamma + \beta\gamma = 3$

iii) $\alpha\beta\gamma = -4$

iv) $\frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} = \frac{2}{-4}$

$$= -\frac{1}{2}$$

2. a)

$$\frac{1}{3}$$

b)

$$4x^2 = 7x + 11$$

$$4x^2 - 7x - 11 = 0$$

$$(4x - 11)(x + 1) = 0$$

$$x = \frac{11}{4}, -1$$

but -1 does not satisfy original equation

$$\therefore x = \frac{11}{4}$$

c)

$$\cos 2\theta = 2\cos^2\theta - 1$$

$$\frac{3}{4} = 2\cos^2\theta - 1$$

$$\cos^2\theta = \frac{7}{8}$$

$$\cos\theta = \sqrt{\frac{7}{8}} \quad (\text{true as } \theta \text{ is acute})$$

d)

$$p(-2) = -16 + 44 - 34 + 6$$

$$= 0$$

$\therefore (x+2)$ is a factor

$$\therefore (x+2)(2x^2 + 7x + 3)$$

$$= (x+2)(2x+1)(x+3)$$

e)

$$-b \leq x-a \leq b$$

$$a-b \leq x \leq a+b$$

$$\therefore a-b=2$$

$$a+b=5$$

$$\therefore a = 3\frac{1}{2}, b = 1\frac{1}{2}$$

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$$3. \quad a) \quad 2x^2 + 3x - 4 \equiv A(x-1)^2 + Bx(x-2) + C(x-1)$$

$$\text{let } x=1: \quad 1 = -B \quad \Rightarrow \quad B = -1$$

$$\text{look at } x^2: \quad 2 = A + B \quad \Rightarrow \quad A = 3$$

$$\text{let } x=0: \quad -4 = A - C \quad \Rightarrow \quad C = 7$$

b) parallel $\Rightarrow$ equal gradients

$$-\frac{1}{2} = \frac{-(2+k)}{3k-1}$$

$$3k-1 = 4+2k$$

$$k = 5$$

c)

$$2 \sin x - 5 \cos x = 2$$

$$2 \left(\frac{2t}{1+t^2} \right) - 5 \left(\frac{1-t^2}{1+t^2} \right) = 2$$

$$4t - 5 + 5t^2 = 2 + 2t^2$$

$$3t^2 + 4t - 7 = 0$$

$$(3t+7)(t-1) = 0$$

$$t = -\frac{7}{3}, 1$$

$$\tan \frac{x}{2} = -\frac{7}{3}, 1$$

$$\frac{x}{2} = 113.2^\circ, 45^\circ$$

$$x = 226^\circ, 90^\circ$$

d)

$$i) \quad m = \frac{ap^2 - aq^2}{2ap - 2aq}$$

$$= \frac{a(p-q)(p+q)}{2a(p-q)}$$

$$= \frac{p+q}{2}$$

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$$ii) \quad y - ap^2 = \frac{p+q}{2} (x - 2ap)$$

$$y - ap^2 = \left(\frac{p+q}{2}\right)x - ap^2 - apq$$

$$y = \left(\frac{p+q}{2}\right)x - apq$$

$$4. \quad a) \quad y = \frac{1}{8}x^2 + x$$

$$8y = x^2 + 8x \quad (1)$$

$$8y + 16 = (x+4)^2$$

$$4 \times 2(y+2) = (x+4)^2$$

$$\therefore \text{Vertex is } (-4, -2)$$

$$\text{focus is } (-4, 0)$$

b)

$$\begin{array}{r} x \\ x^2 - 1 \overline{) x^3 + 4x - 4} \\ \underline{x^3 - x} \\ 5x - 4 \end{array} \quad (1)$$

$$\therefore \text{remainder} = 5x - 4$$

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c) Solve simultaneously (1 solution)

$$x^2 - 2x = kx - 16$$

$$x^2 - 2x - kx + 16 = 0$$

$$x^2 - (2+k)x + 16 = 0$$

1 solution $\Rightarrow \Delta = 0$

$$(2+k)^2 - 4 \times 16 = 0$$

$$k^2 + 4k - 60 = 0$$

$$(k+10)(k-6) = 0$$

$$k = -10, 6$$

d)

$$2 \sin 2\theta = 3 \tan \theta$$

$$4 \sin \theta \cos \theta = \frac{3 \sin \theta}{\cos \theta}$$

$$4 \sin \theta \cos^2 \theta - 3 \sin \theta = 0$$

$$\sin \theta (4 \cos^2 \theta - 3) = 0$$

$$\sin \theta = 0 \quad \cos \theta = \pm \frac{\sqrt{3}}{2}$$

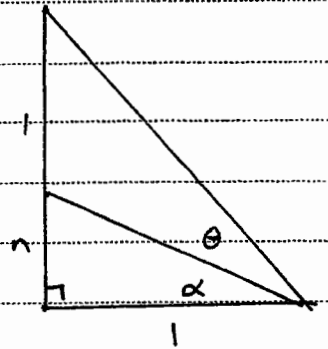
$$\theta = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

Teacher's Name: _____

Student's Name/N^o: _____

5. a) $\tan \alpha = n$

$\tan (\theta + \alpha) = n + 1$



$$\tan (\theta + \alpha) = n + 1$$

$$\frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = n + 1$$

$$\frac{\tan \theta + n}{1 - n \tan \theta} = n + 1$$

$$\tan \theta + n = n + 1 - n^2 \tan \theta - n \tan \theta$$

$$\tan \theta + n^2 \tan \theta + n \tan \theta = 1$$

$$\tan \theta = \frac{1}{n^2 + n + 1}$$

b) i) $y = \frac{x^2}{8}$

$$y' = \frac{x}{4} \quad \text{when } x = 4p$$

$$m_T = \frac{4p}{4}$$

$$= p$$

$$\therefore y - 2p^2 = p(x - 4p)$$

$$y - 2p^2 = px - 4p^2$$

$$y = px - 2p^2$$

ii) when $x = 0$

$$y = -2p^2$$

$$\therefore T(0, -2p^2)$$

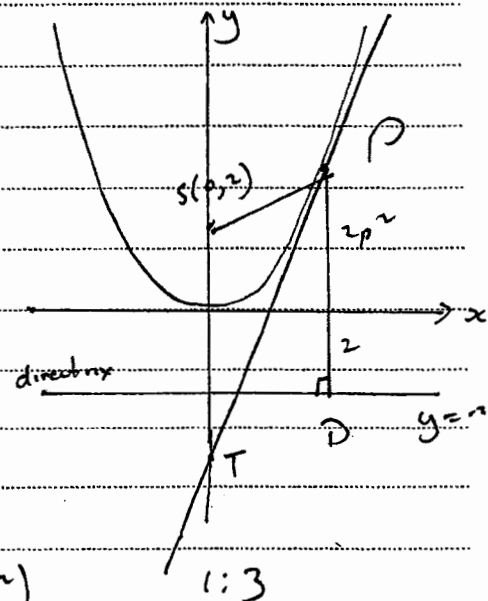
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iii)

$$TS = 2 + 2p^2$$

$$\begin{aligned} PS &= PD \text{ (by definition)} \\ &= 2 + 2p^2 \\ &= TS \end{aligned}$$



$$iv) P(4p, 2p^2) \quad T(0, -2p^2)$$

1:3

$$\therefore M \left(\frac{3 \times 4p + 0}{4}, \frac{3 \times 2p^2 + 1 \times -2p^2}{4} \right)$$

$$M(3p, p^2)$$

$$\therefore x = 3p, \quad y = p^2$$

$$\therefore p = \frac{x}{3}$$

$$\therefore y = \frac{x^2}{9} \text{ is the locus.}$$

6. a) i) $PA = 2 \times PB$

$$\sqrt{(x-5)^2 + (y-6)^2} = 2 \times \sqrt{(x+1)^2 + y^2}$$

$$x^2 - 10x + 25 + y^2 - 12y + 36 = 4[x^2 + 2x + 1 + y^2]$$

$$3x^2 + 18x + 3y^2 + 12y = 57$$

$$x^2 + 6x + y^2 + 4y = 19$$

which is a circle

ii) $x^2 + 6x + 9 + y^2 + 4y + 4 = 19 + 9 + 4$

$$(x+3)^2 + (y+2)^2 = 32$$

$\therefore$ Centre $(-3, -2)$

radius $\sqrt{32}$ units

b) let roots be $\alpha, \beta, \alpha + \beta$

i) $\alpha + \beta + \alpha + \beta = -6$ (sum of roots)

$$\alpha + \beta = -3$$

$\therefore -3$ is a root

sub into equation

$$(-3)^3 + 6(-3)^2 - (-3) + m = 0$$

$$m = -30$$

ii) $x^3 + 6x^2 - x - 30 = 0$

has -3 as a root

$\therefore (x+3)$ is a factor

$\therefore (x+3)(x^2 + 3x - 10) = 0$ by inspection

$$(x+3)(x+5)(x-2) = 0$$

$$\therefore x = -3, -5, 2$$

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$$c) \quad \frac{\tan 2\theta - \tan \theta}{\cot \theta + \tan 2\theta}$$

$$= \frac{\frac{2\tan \theta}{1-\tan^2 \theta} - \tan \theta}{\frac{1}{\tan \theta} + \frac{2\tan \theta}{1-\tan^2 \theta}}$$

$$\frac{1}{\tan \theta} + \frac{2\tan \theta}{1-\tan^2 \theta}$$

$$= \frac{2\tan^2 \theta - \tan^2 \theta (1-\tan^2 \theta)}{1-\tan^2 \theta + 2\tan^2 \theta}$$

$$= \frac{\tan^2 \theta + \tan^4 \theta}{1+\tan^2 \theta}$$

$$= \tan^2 \theta$$